2025-DSE
PHY

PAPER 2

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HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2025

Candidate Number									

PHYSICS PAPER 2

Question-Answer Book

11:45 am – 12:45 pm (1 hour)
This paper must be answered in English

INSTRUCTIONS

(1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick barcode labels in the spaces provided on Pages 1, 3, 5, 7 and 9.

(2) This paper consists of **FOUR** sections, Sections A, B, C and D. Each section contains eight multiple-choice questions and one structured question which carries 10 marks. Attempt **ALL** questions in any **TWO** sections.

(3) Write your answers to the structured questions in the **ANSWER BOOK** provided. For multiple-choice questions, blacken the appropriate circle with an HB pencil. You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

(4) Graph paper and supplementary answer sheets will be provided on request. Write your candidate number, mark the question number box and stick a barcode label on each sheet, and fasten them with string **INSIDE** the Answer Book.

(5) The Question-Answer Book and Answer Book will be collected **SEPARATELY** at the end of the examination.

(6) The diagrams in this paper are **NOT** necessarily drawn to scale.

(7) The last two pages of this Question-Answer Book contain a list of data, formulae and relationships which you may find useful.

(8) No extra time will be given to candidates for sticking on the barcode labels or filling in the question number boxes after the 'Time is up' announcement.

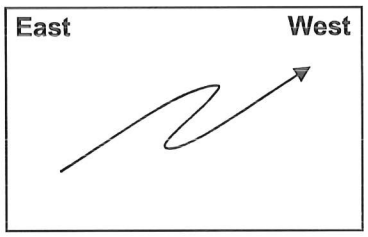
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Section A : Astronomy and Space Science

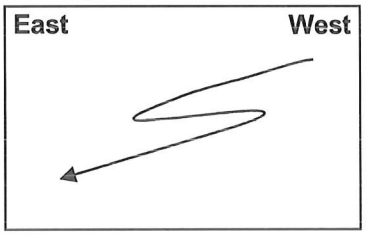
Q.1: Multiple-choice questions

1.1 Which of the following paths of a planet viewed against distant background stars show(s) retrograde motion ?

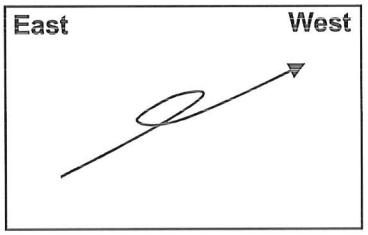
(1)



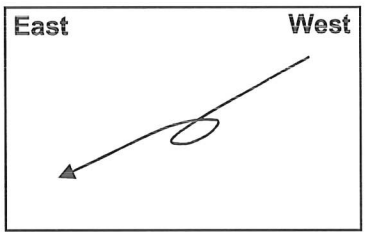
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(3)



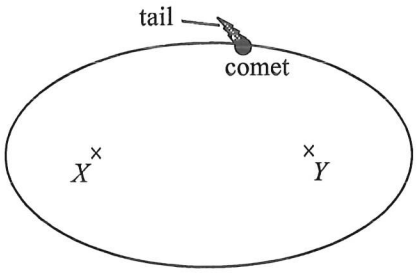
(4)



A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (4) only

A	B	C	D
○	○	○	○

1.2 A comet revolving about the Sun takes an elliptical orbit with foci X and Y as shown. It is known that the tail of a comet is always directed away from the Sun.



Which of the following descriptions about the comet **MUST** be correct ?

(1) The comet is travelling in a clockwise direction.
(2) The comet attains its greatest speed when it is nearest to Y.
(3) The gravitational potential energy of the comet in the orbit remains unchanged.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

A	B	C	D
○	○	○	○

Please stick the barcode label here.

1.3 Star Y is measured to have a parallax angle of p from telescopes on the Earth's surface. Suppose star Y is measured from telescopes on the surface of Mars (orbital radius 1.5 AU), the parallax angle would be

A. larger than p.
B. equal to p.
C. smaller than p.
D. either smaller, larger or equal to p.

A ○ B ○ C ○ D ○

1.4

star	luminosity	intensity observed	colour
X	5	3	white
Y	2	4	orange
Z	3	1	blue

The table shows some properties of stars X, Y and Z. The luminosity and intensity observed are given respectively in the same arbitrary units. Which star is (1) the hottest; and (2) seen to be the brightest ?

	(1) hottest	(2) brightest
A.	Y	Z
B.	Z	Y
C.	Z	X
D.	X	Y

A ○ B ○ C ○ D ○

1.5 Stars X and Y have the same surface temperature. The luminosities of X and Y are $L_X = 121000\ L_S$ and $L_Y = 0.1\ L_S$ respectively, where L_S is the luminosity of the Sun. Find $R_X : R_Y$, where R_X and R_Y are radii of X and Y respectively.

A. 1210000 : 1
B. 1 : 1210000
C. 1100 : 1
D. 1 : 1100

A ○ B ○ C ○ D ○

1.6 The spectral class of a main-sequence star can be determined by

(1) the surface temperature of the star.
(2) the colour of the star.
(3) the intensity of the lines in the stellar spectrum.

A. (1), (2) and (3)
B. (1) and (3) only
C. (2) only
D. (1) only

A ○ B ○ C ○ D ○

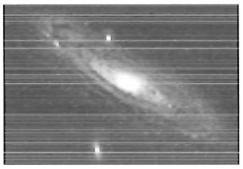
1.7 In the Hertzsprung-Russell (H-R) diagram, stars of the same apparent magnitude occupy

A. the same horizontal line.
B. the same vertical line.
C. the same diagonal going from lower-right to upper-left.
D. arbitrary locations in the diagram.

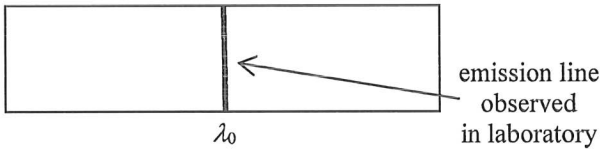
A	B	C	D
○	○	○	○

1.8

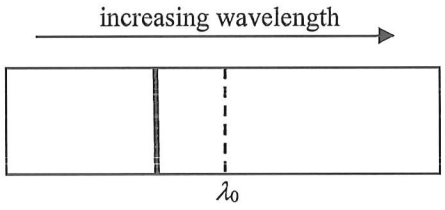




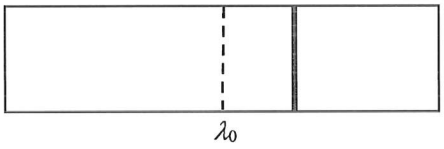
Which of the following best shows the observed emission line (rest wavelength λ_0 observed in laboratory) of the above blueshifted spiral galaxy ?



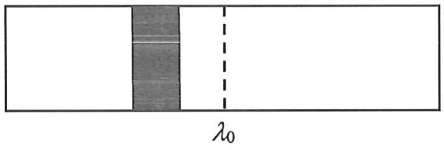
A.



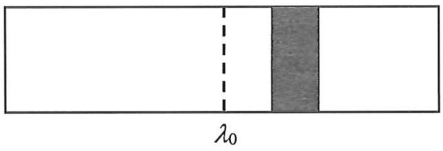
B.



C.



D.



A	B	C	D
○	○	○	○

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Q.1: Structured question

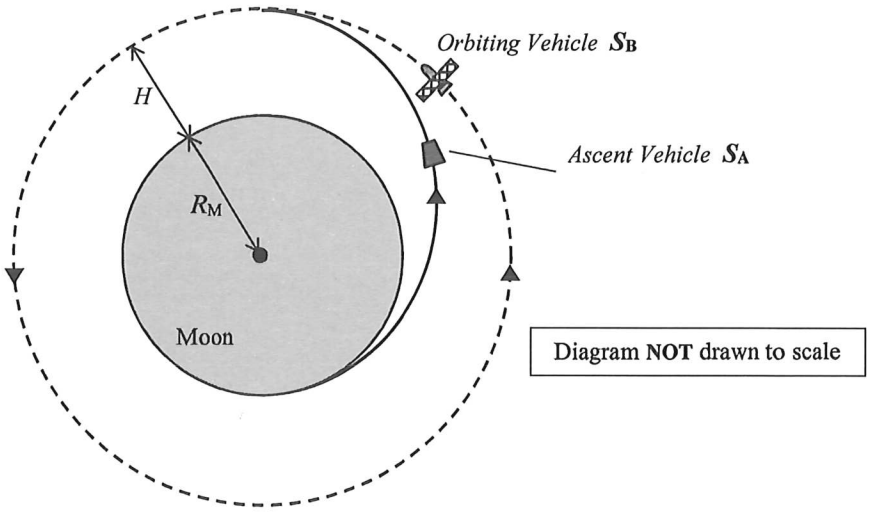
Given: mass of the Moon $M_M = 7.34 \times 10^{22}$ kg
radius of the Moon $R_M = 1.74 \times 10^6$ m

To set up a permanent lunar station on the Moon, one has to overcome the issue of transporting astronauts/supplies between the Earth and the lunar station.

The following two-stage return scheme is proposed for bringing astronauts from the Moon back to the Earth:

(I) An *Ascent Vehicle* S_A is launched from the Moon's surface to dock with an *Orbiting Vehicle* S_B which is orbiting circularly around the Moon at an altitude H above the Moon's surface.
(II) The *Orbiting Vehicle* carrying the astronauts then returns to the Earth.

Figure 1.1



You may neglect the gravitational effects of the Earth in your calculation. Given: $H = 0.20 \times 10^6$ m

(a) Find the orbital speed v_o of S_B, which is close to 1.6 km s^{-1}. Give your answer correct to 3 significant figures. (3 marks)

(b) An astronaut inside S_B experiences weightlessness. A student suggests that this is because the Moon's gravitational force acting on the astronaut is zero. Explain whether this is correct or not. (2 marks)

In planning the return journey, the *Ascent Vehicle* S_A launched at velocity v_1 from the Moon's surface will then dock with the orbiting S_B, where the two space vehicles are moving with the same speed v_o (i.e. the orbital speed of S_B) when docking.

(c) (i) Determine the launching velocity v_1 required. Assume that the rocket engine of S_A is shut immediately after it has attained v_1 on the Moon's surface. (3 marks)

(ii) Referring to the escape velocity $v_e = 2.38$ km s^{-1} of the Moon, explain, regarding the design of S_A, an advantage of this two-stage return scheme compared to a direct return flight. (2 marks)

Section B : Atomic World

Q.2: Multiple-choice questions

2.1 In Thomson's plum pudding model, it is suggested that an atom is a positively charged sphere with electrons distributed evenly within it. Which of the following results of the α-scattering experiment implies that Thomson's model is **incorrect** ?

(1) Most of the α particles pass straight through the thin gold foil.
(2) Some α particles are deflected by large angles.
(3) A few α particles are bounced back.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

A	B	C	D
○	○	○	○

2.2 Which of the following about the Bohr atomic model is/are correct ?

(1) Electrons are orbiting around the nucleus.
(2) The attractive force between the nucleus and the electron provides the centripetal force for the electron's circular motion.
(3) Photons are emitted when electrons jump from lower energy orbits to higher energy orbits.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

A	B	C	D
○	○	○	○

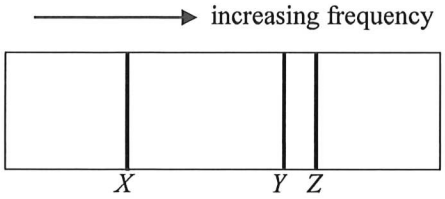
2.3 The energy levels of a hydrogen-like atom take the form $-\frac{36}{n^2}$ eV, where n is the quantum number. What is the energy level of its 2nd excited state ?

A. –4 eV
B. –9 eV
C. –12 eV
D. –18 eV

A	B	C	D
○	○	○	○

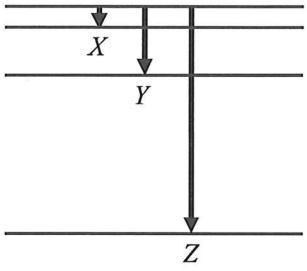
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2.4

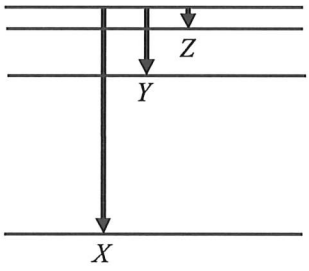


The above figure shows the spectrum obtained when the visible light from a discharge tube passes through a diffraction grating. Spectral lines *X*, *Y* and *Z* correspond to electron transitions between four energy levels. Which of the following best illustrates the electron transitions that give the spectral lines ?

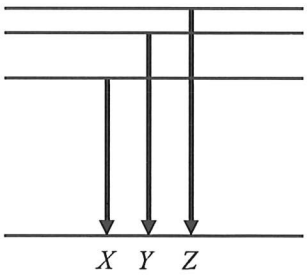
A.



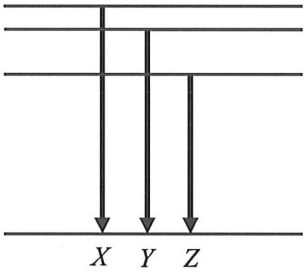
B.



C.



D.



A B C D

○ ○ ○ ○

2.5 A beam of fast-moving electrons of velocity *v* behaves like a wave with wavelength λ and frequency *f*. Which of the following statements is/are correct ?

(1) λ increases when *v* decreases.

(2) The beam of electrons shows wave properties only with some discrete *f*.

(3) The momentum associated with the electron beam is given by $\frac{\lambda}{\text{Planck constant}}$.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

A B C D

○ ○ ○ ○

2.6 Two point objects of separation 2.4×10^{-6} m are to be observed under a microscope using light of wavelength 550 nm. Which of the following combinations of aperture diameter and focal length of the microscope's objective lens is suitable ?

	aperture diameter	**focal length**
A.	5 mm	20 mm
B.	6 mm	20 mm
C.	5 mm	24 mm
D.	6 mm	24 mm

A	B	C	D
○	○	○	○

2.7 Which of the following statements about a scanning tunnelling microscope (STM) is/are correct ?

(1) It uses a high energy electron beam instead of visible light.
(2) STM gives a 3-dimentional image of a conductive surface.
(3) The tunnelling current produced depends on the separation between the surface under investigation and the probe of the STM.

		A	B	C	D
A.	(1) only	○	○	○	○
B.	(3) only				
C.	(1) and (2) only				
D.	(2) and (3) only				

2.8 When a material is broken up into nanosized particles, which of the following statements is/are correct ?

(1) Its chemical properties change but the physical properties remain unchanged.
(2) Its shape changes but the atomic arrangement remains unchanged.
(3) Its total surface area to volume ratio increases.

		A	B	C	D
A.	(1) only	○	○	○	○
B.	(3) only				
C.	(2) and (3) only				
D.	(1), (2) and (3)				

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Q.2: Structured question

(a) Figure 2.1 shows the lowest four energy levels of a hydrogen atom.

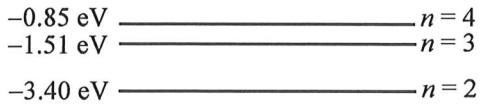


Figure 2.1



When a hydrogen atom in ground state and a photon of energy 10.2 eV collide,

(i) state all subsequent changes to the hydrogen atom. (2 marks)

(ii) find the wavelength of the photon subsequently emitted by the hydrogen atom. (2 marks)

(b) The electrode X of the photocell shown below is made of zinc and it is illuminated by photons of energy 10.2 eV. Given: work function of zinc is 4.9 eV

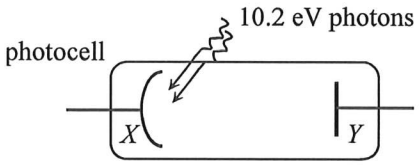


Figure 2.2

(i) Find the maximum speed v_{max} of the photoelectrons emitted from electrode X. (2 marks)

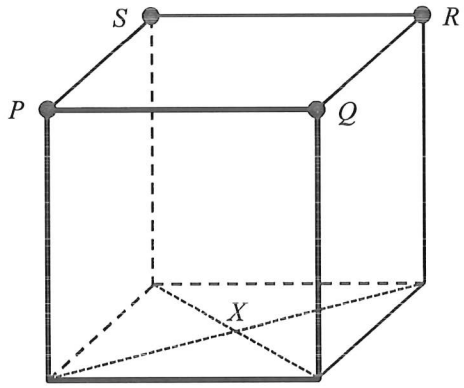
(ii) In your answer book, sketch a graph (including the origin) showing how this maximum speed v_{max} depends on the energy E_p of the incident photons. (2 marks)

(c) Explain why 10.2 eV photons are able to remove electrons from the zinc surface whereas they can only excite hydrogen atoms in ground state without releasing the electron from the atom. (2 marks)

Section C : Energy and Use of Energy

Q.3: Multiple-choice questions

3.1 In a room of dimensions $d \times d \times d$, identical point light sources, P, Q, R and S, are located at the four corners of the ceiling as shown. The luminous flux of each source is Φ.



Find the illuminance around point X on the floor in the middle of the room. Neglect reflections from the walls of the room.

A. $\frac{\Phi}{\pi d^2} \times (\frac{3}{2})^{\frac{1}{2}}$

B. $\frac{\Phi}{\pi d^2} \times (\frac{3}{2})^{\frac{3}{2}}$

C. $\frac{\Phi}{\pi d^2} \times (\frac{2}{3})^{\frac{1}{2}}$

D. $\frac{\Phi}{\pi d^2} \times (\frac{2}{3})^{\frac{3}{2}}$

A	B	C	D
○	○	○	○

3.2 The end-use energy efficiency of microwave ovens is high because

(1) microwaves only heat up the food without heating up the utensil directly.
(2) microwaves do not heat up the nearby air much.
(3) no energy is lost as heat in the circuit of the microwaves-generating component.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

A	B	C	D
○	○	○	○

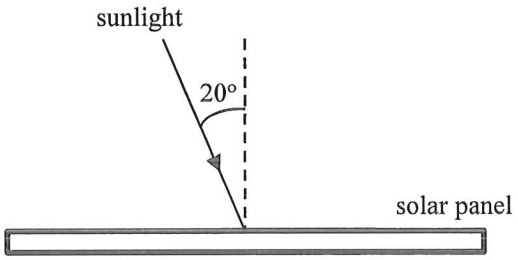
3.3 Which of the following arrangements can reduce the Overall Thermal Transfer Value (OTTV) of a building envelope ?

(1) using air-conditioners which are more energy-saving
(2) using glass windows with low-emissivity coating
(3) using building materials with greater U-value

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

A	B	C	D
○	○	○	○

3.4 When incident sunlight makes an angle of 20° to the normal of a solar panel, the electric power generated is 100 W.



Assuming that the efficiency of the solar panel remains unchanged, find the electric power generated if the sunlight makes an angle of 30° to the normal of the panel and its intensity is reduced by 10%.

A. $100 \times \frac{\sin 30^o}{\sin 20^o} \times 0.1$ W

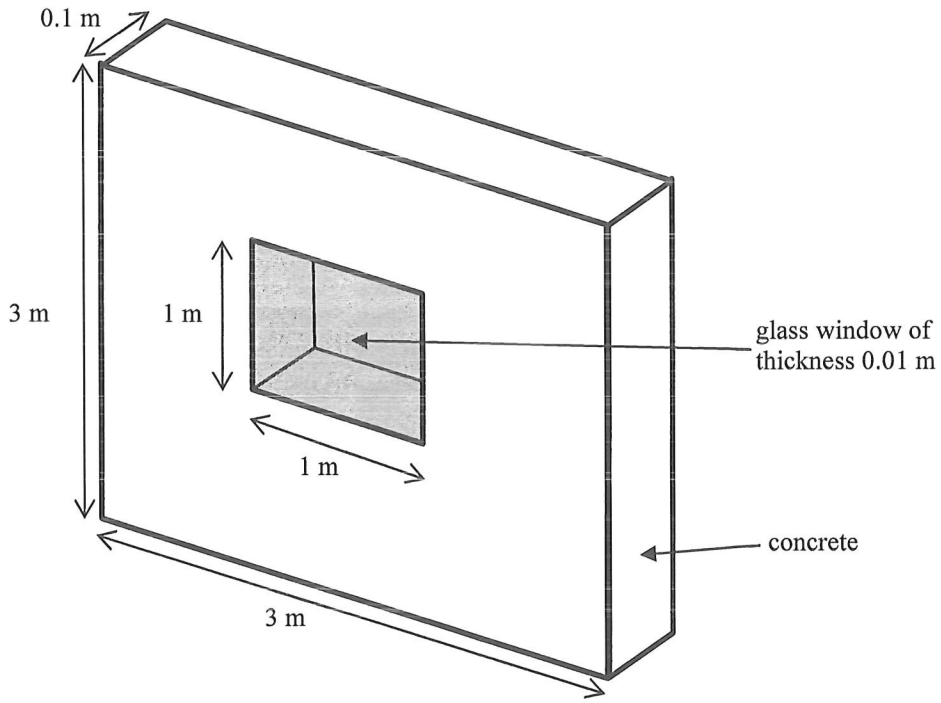
B. $100 \times \frac{\cos 30^o}{\cos 20^o} \times 0.1$ W

C. $100 \times \frac{\sin 30^o}{\sin 20^o} \times 0.9$ W

D. $100 \times \frac{\cos 30^o}{\cos 20^o} \times 0.9$ W

A	B	C	D
○	○	○	○

3.5 The figure shows a 3 m × 3 m wall made of concrete of thickness 0.1 m and having a 1 m × 1 m glass window of thickness 0.01 m.



The thermal conductivities of concrete and glass are 0.8 W m^{-1} K^{-1} and 1.3 W m^{-1} K^{-1} respectively. Find the overall thermal transmittance (U-value) of the wall.

A. 20 W m^{-2} K^{-1}

B. 22 W m^{-2} K^{-1}

C. 73 W m^{-2} K^{-1}

D. 116 W m^{-2} K^{-1}

A	B	C	D
○	○	○	○

3.6 Air-conditioners P and Q below operate respectively to cool two identical rooms under the same environmental conditions.

air-conditioner	input power / kW	coefficient of performance (COP)
P	1.6	2.2
Q	2.0	1.9

If P takes 400 s to cool down the room, estimate the time required for Q to cool down the other room to the same extent.

A. 688 s
B. 432 s
C. 371 s
D. 276 s

A	B	C	D
○	○	○	○

3.7 Wind turbines P and Q below deliver roughly the same electric power output when they operate under the conditions tabulated.

wind turbine	blade length / m	wind speed / m s^{-1} (blowing normally to the turbine)
P	23	6.2
Q	15	8.0

The overall efficiency of electric power generation

A. is roughly the same for both turbines.
B. is larger for P.
C. is larger for Q.
D. of the turbines cannot be compared as the maximum power available for tapping from the wind is not the same.

A	B	C	D
○	○	○	○

3.8 What are the functions of pressurised water in the core of a pressurised water reactor ?

(1) It acts as a moderator.
(2) It acts as a coolant.
(3) It turns into steam and drives the turbine.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

A	B	C	D
○	○	○	○

Q.3: Structured question

Indoor vertical farming is suitable for urban areas where land is limited. Figure 3.1 shows a container modified into a vertical farming system for growing plants under optimal conditions provided by artificial lighting and temperature control.

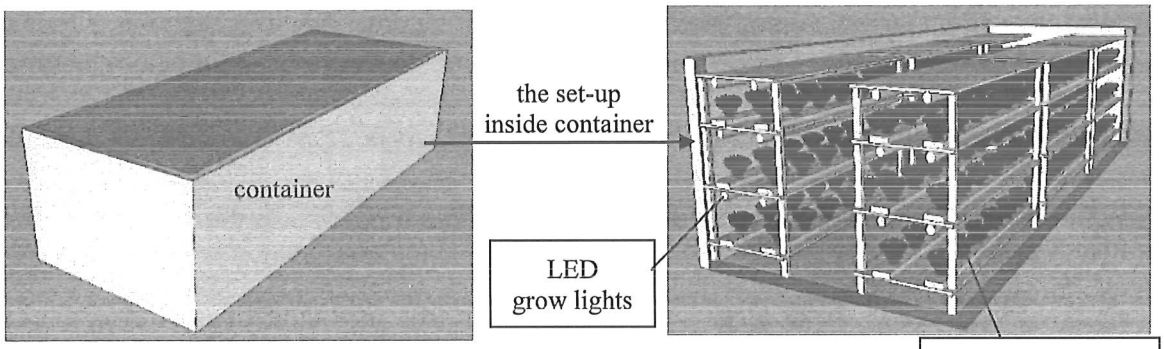


Figure 3.1

(a) (i) State **TWO** reasons why light emitting diodes (LEDs) are preferred to be used as grow lights rather than fluorescent tube lamps (FTLs). (2 marks)

(ii) Taking into account energy efficiency, state which lighting system is the best for growing green leaf plants: (I) green LEDs, (II) red and blue LEDs and (III) full spectrum (white) LEDs. (1 mark)

(b) Assume that all surfaces of the container are made from the same metal plate and that heat transfer through the floor is negligible. The container's interior temperature is maintained at an optimum value of 20 °C. The table below shows the variation of the interior and exterior temperatures of the container within 24 hours.

time	LEDs lighting system	interior temperature (°C)	exterior temperature (°C)
8:00 a.m. - 8:00 p.m.	ON	20	32
8:00 p.m. - 8:00 a.m.	OFF	20	24

Given: total exposed surface area of the container = 70 m^2
U-value of the metal plate of the container = 12.5 W m^{-2} K^{-1}

(i) Estimate the total thermal energy gain from outside the container during this 24-hour period. (2 marks)

(ii) Given that the air-conditioning system consumes 0.3 J of electrical energy for removing 1 J of thermal energy from the container. Estimate the cost of electricity per day if the air-conditioning system is required to remove 6.45×10^8 J of thermal energy during this 24-hour period.
Given : cost of electricity = $1.4 per kW h (2 marks)

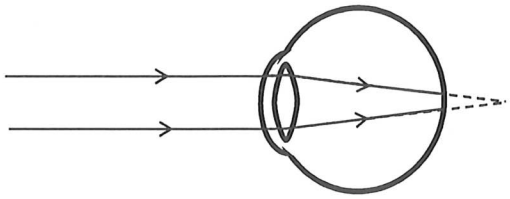
(iii) Estimate the minimum cooling capacity of the container's air-conditioning system, in kW, required for maintaining the interior temperature at 20 °C. Given that the total power of the lighting system is 1500 W and 60% of this input power eventually becomes thermal energy. (2 marks)

(c) Without changing the dimensions and material of the container, suggest one modification to the container so as to reduce the energy consumption. (1 mark)

Section D : Medical Physics

Q.4: Multiple-choice questions

4.1 The figure shows an eye viewing a distant object.



Which of the following descriptions is correct ?

	eye lens	**can be corrected by**
A.	too thin	a converging lens
B.	too thick	a converging lens
C.	too thin	a diverging lens
D.	too thick	a diverging lens

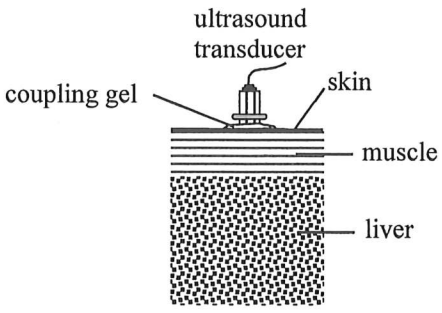
A	B	C	D
○	○	○	○

4.2 The human eye is most sensitive to light of wavelength 550 nm. Given that the diameter of the pupil of a normal eye is 5 mm, estimate the angular resolving power of the eye.

A. 0.0008°
B. 0.008°
C. 0.08°
D. 0.8°

A	B	C	D
○	○	○	○

4.3 An ultrasound transducer is set up as shown below.



The acoustic impedance of various tissues and the coupling gel used are given below.

	acoustic impedance / $\times 10^6$ kg m^{-2} s^{-1}
coupling gel	1.45
muscle	1.71
liver	1.65

Estimate the proportion of the intensity of the incident ultrasound beam that enters the liver from the coupling gel.

A. 0.9929
B. 0.9932
C. 0.9997
D. 0.0071

A	B	C	D
○	○	○	○

4.4 Which of the following statements about ultrasound scan is/are correct ?

(1) Performing an ultrasound scan does not require great skills and experience.
(2) Ultrasound cannot penetrate bone, so it is difficult to image the human brain.
(3) Lung imaging is difficult as there are air spaces in the lung.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

A	B	C	D
○	○	○	○

4.5 Which of the following statements about X-ray radiographic imaging is/are correct ?

(1) A radiation dose is cumulative.
(2) The resolution of soft tissues is good using high energy X-rays.
(3) The use of contrast media can be unpleasant.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

A	B	C	D
○	○	○	○

4.6 The half-value thickness of a cobalt-60 γsource (1 MeV) is 12.5 mm in lead and 60.5 mm in concrete. Estimate the minimum thickness of a lead sheet to replace a concrete wall of 25.0 cm thick for radiation shielding of this cobalt-60 source.

A. 6.1 cm
B. 5.2 cm
C. 4.0 cm
D. 3.1 cm

A	B	C	D
○	○	○	○

4.7 During a computed tomography (CT) scan, an X-ray tube rotates around the patient and emits X-rays while detectors collect the intensity of X-ray transmission through the patient. Which of the following is the correct sequence of further processing ?

(1) Attenuation data are put through a reconstruction algorithm.
(2) The intensity of X-ray transmission is converted to digital data.
(3) Digital data are processed to obtain attenuation data.
(4) The reconstructed image is available for viewing on a digital display.

A. (3), (1), (4), (2)
B. (2), (4), (3), (1)
C. (3), (1), (2), (4)
D. (2), (3), (1), (4)

A	B	C	D
○	○	○	○

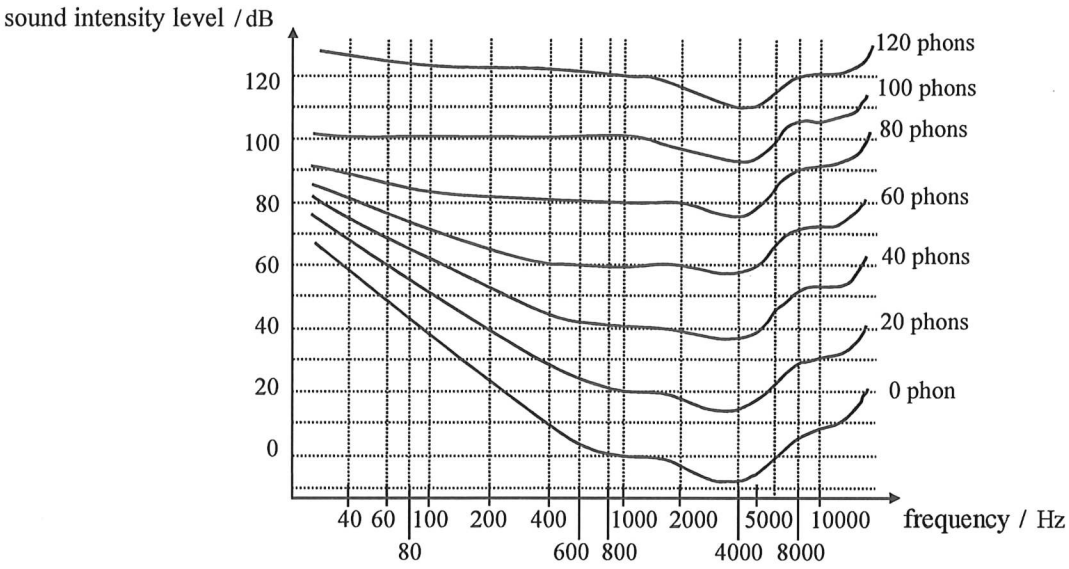
4.8 A tracer has a biological half-life of 9.45 days and an effective half-life of 2.30 days. Find its physical half-life.

A. 1.85 days
B. 7.15 days
C. 3.04 days
D. 11.8 days

A	B	C	D
○	○	○	○

Q.4: Structured question

The figure shows the curves of equal loudness in the phon scale for persons with normal hearing.



(a) (i) Explain why they are called 'curves of equal loudness'. (1 mark)

(ii) What is the physical significance of the 0 phon curve ? (1 mark)

(b) (i) Which of the following in the ear – **the three ossicles, the hair sensory cells** and **the auditory nerve** - is responsible for the generation of nerve impulse for hearing ? (1 mark)

(ii) From what feature of these curves does one know that the ear is most sensitive to sound between 2000 Hz and 5000 Hz ? (1 mark)

(c) (i) At what frequency are loudness and the sound intensity level numerically the same ? (1 mark)

(ii) When the loudness changes from 20 phons to 40 phons, determine whether the change of sound intensity level (in dB) at 60 Hz is the same as that at 8 kHz. Show your working. (2 marks)

(d) (i) Sounds above 120 phons are painful as well as damaging. Calculate the ratio of intensities of sound between 120 phons and 0 phon at 1000 Hz. (2 marks)

(ii) How would these curves change for people who suffer from hearing loss mostly for sounds of high frequencies ? (1 mark)

END OF PAPER

Sources of materials used in this paper will be acknowledged in the *HKDSE Question Papers* booklet published by the Hong Kong Examinations and Assessment Authority at a later stage.

Do not write on this page.

Answers written on this page will not be marked.

List of data, formulae and relationships

Data

molar gas constant	$R = 8.31$ J mol^{-1} K^{-1}	
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}	
acceleration due to gravity	$g = 9.81$ m s^{-2} (close to the Earth)	
universal gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}	
speed of light in vacuum	$c = 3.00 \times 10^8$ m s^{-1}	
charge of electron	$q_e = 1.60 \times 10^{-19}$ C	
electron rest mass	$m_e = 9.11 \times 10^{-31}$ kg	
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}$ C^2 N^{-1} m^{-2}	
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}	
atomic mass unit	u $= 1.661 \times 10^{-27}$ kg	(1 u is equivalent to 931 MeV)
astronomical unit	AU $= 1.50 \times 10^{11}$ m	
light year	ly $= 9.46 \times 10^{15}$ m	
parsec	pc $= 3.09 \times 10^{16}$ m $= 3.26$ ly $= 206265$ AU	
Stefan constant	$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}	
Planck constant	$h = 6.63 \times 10^{-34}$ J s	

Rectilinear motion

For uniformly accelerated motion :

$$v = u + at$$
$$s = ut + \frac{1}{2}at^2$$
$$v^2 = u^2 + 2as$$

Mathematics

Equation of a straight line	$y = mx + c$
Arc length	$= r\theta$
Surface area of cylinder	$= 2\pi rh + 2\pi r^2$
Volume of cylinder	$= \pi r^2 h$
Surface area of sphere	$= 4\pi r^2$
Volume of sphere	$= \frac{4}{3}\pi r^3$

For small angles, $\sin\theta \approx \tan\theta \approx \theta$ (in radians)

Astronomy and Space Science	
$U = -\frac{GMm}{r}$	gravitational potential energy
$P = \sigma AT^4$	Stefan's law
$\left\|\frac{\Delta f}{f_0}\right\| \approx \frac{v}{c} \approx \left\|\frac{\Delta\lambda}{\lambda_0}\right\|$	Doppler effect

Energy and Use of Energy	
$E = \frac{\Phi}{A}$	illuminance
$\frac{Q}{t} = \kappa\frac{A(T_H - T_C)}{d}$	rate of energy transfer by conduction
$U = \frac{\kappa}{d}$	thermal transmittance U-value
$P = \frac{1}{2}\rho A v^3$	maximum power by wind turbine

Atomic World	
$\frac{1}{2}m_e v_{max}{}^2 = hf - \phi$	Einstein's photoelectric equation
$E_n = -\frac{1}{n^2}\left\{\frac{m_e q_e^4}{8h^2\varepsilon_0^2}\right\} = -\frac{13.6}{n^2}$ eV	energy level equation for hydrogen atom
$\lambda = \frac{h}{p} = \frac{h}{mv}$	de Broglie formula
$\theta \approx \frac{1.22\lambda}{d}$	Rayleigh criterion (resolving power)

Medical Physics	
$\theta \approx \frac{1.22\lambda}{d}$	Rayleigh criterion (resolving power)
power $= \frac{1}{f}$	power of a lens
$L = 10\log\frac{I}{I_0}$	intensity level (dB)
$Z = \rho c$	acoustic impedance
$\alpha = \frac{I_r}{I_0} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$	intensity reflection coefficient
$I = I_0 e^{-\mu x}$	transmitted intensity through a medium

A1.	$E = mc\,\Delta T$	energy transfer during heating and cooling
A2.	$E = l\,\Delta m$	energy transfer during change of state
A3.	$pV = nRT$	equation of state for an ideal gas
A4.	$pV = \frac{1}{3}Nm\overline{c^2}$	kinetic theory equation
A5.	$E_K = \frac{3RT}{2N_A}$	molecular kinetic energy

B1.	$F = m\frac{\Delta v}{\Delta t} = \frac{\Delta p}{\Delta t}$	force
B2.	moment $= F \times d$	moment of a force
B3.	$E_P = mgh$	gravitational potential energy
B4.	$E_K = \frac{1}{2}mv^2$	kinetic energy
B5.	$P = Fv$	mechanical power
B6.	$a = \frac{v^2}{r} = \omega^2 r$	centripetal acceleration
B7.	$F = \frac{Gm_1m_2}{r^2}$	Newton's law of gravitation

C1.	$\Delta y = \frac{\lambda D}{a}$	fringe separation in double-slit interference
C2.	$d\sin\theta = n\lambda$	diffraction grating equation
C3.	$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$	equation for a single lens

D1.	$F = \frac{Q_1Q_2}{4\pi\varepsilon_0 r^2}$	Coulomb's law
D2.	$E = \frac{Q}{4\pi\varepsilon_0 r^2}$	electric field strength due to a point charge
D3.	$E = \frac{V}{d}$	electric field between parallel plates (numerically)
D4.	$R = \frac{\rho l}{A}$	resistance and resistivity
D5.	$R = R_1 + R_2$	resistors in series
D6.	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$	resistors in parallel
D7.	$P = IV = I^2R$	power in a circuit
D8.	$F = BQv\sin\theta$	force on a moving charge in a magnetic field
D9.	$F = BIl\sin\theta$	force on a current-carrying conductor in a magnetic field
D10.	$B = \frac{\mu_0 I}{2\pi r}$	magnetic field due to a long straight wire
D11.	$B = \frac{\mu_0 NI}{l}$	magnetic field inside a long solenoid
D12.	$\varepsilon = N\frac{\Delta\Phi}{\Delta t}$	induced e.m.f.
D13.	$\frac{V_s}{V_p} \approx \frac{N_s}{N_p}$	ratio of secondary voltage to primary voltage in a transformer

E1.	$N = N_0e^{-kt}$	law of radioactive decay
E2.	$t_{\frac{1}{2}} = \frac{\ln 2}{k}$	half-life and decay constant
E3.	$A = kN$	activity and the number of undecayed nuclei
E4.	$\Delta E = \Delta mc^2$	mass-energy relationship